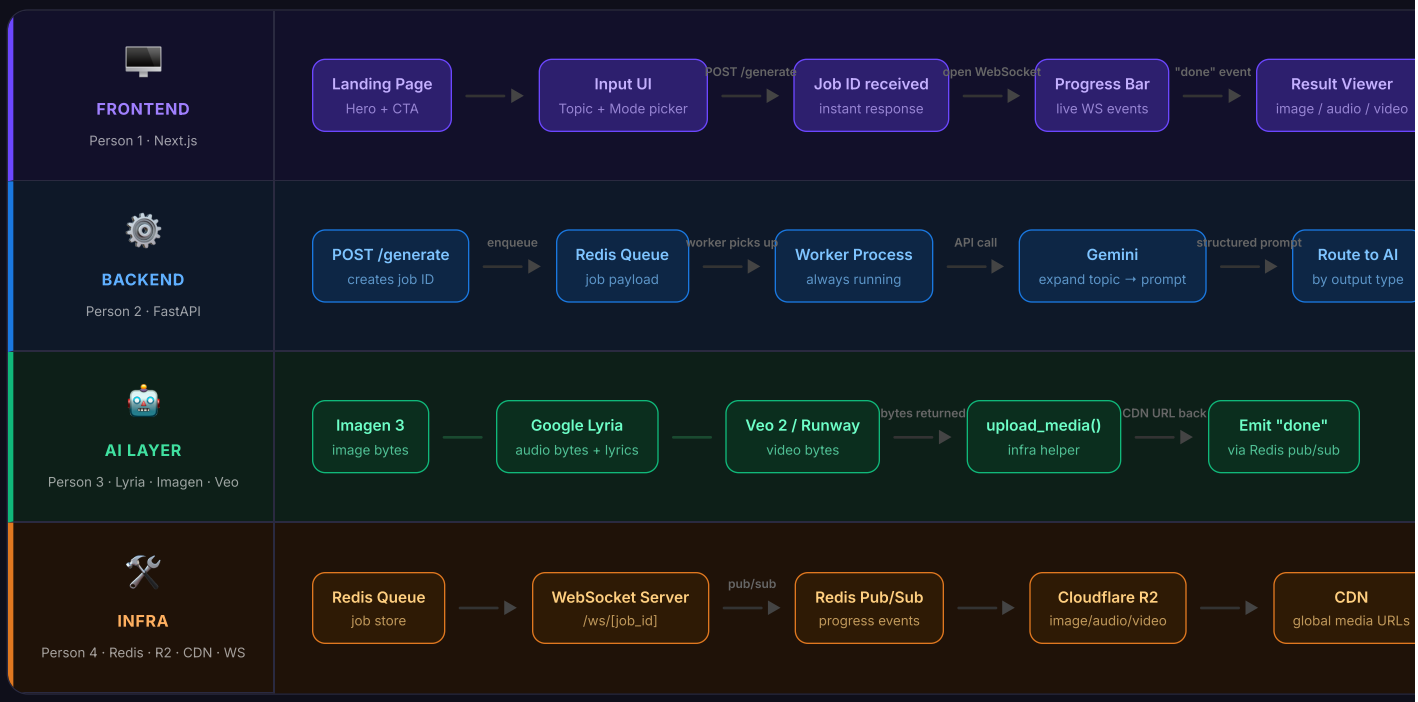
LearnLens — System Architecture

How Frontend, Backend, AI Integrations, and Infrastructure connect at runtime



END-TO-END REQUEST FLOW

Frontend (Next.js · Vercel) Backend (FastAPI · Railway) AI Layer (Imagen · Lyria · Veo) Infra (Redis · R2 · CDN · WebSocket)

Caching shortcut: Before calling any AI API, the worker checks Redis for a cache key `cache:{type}:{topic}`. On a hit, it skips Gemini and all AI calls entirely, fires `done` immediately with the cached CDN URL makes stage presentations instant. | **Share links:** `/s/[job_id]` on the frontend calls `GET /output/[job_id]` on the backend, which reads the CDN URL from Redis and returns it — no re-generation needed.